MMM-A-138B
30 November 1986
SUPERSEDING
MMM-A-138A
2 March 1967

FEDERAL SPECIFICATION

ADHESIVE, METAL TO WOOD, STRUCTURAL

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies

1. SCOPE AND CLASSIFICATION

1 1 Scope This specification covers the requirements for synthetic resin adhesives to structurally bond clad aluminum alloy to wood

1 2 Classification

1 2 1 Types - The adhesives covered by this specification shall be furnished in the following types, as specified

Type I - Single adhesive system
Type II - Two adhesive system

1 2 2 Conditions - The Type II adhesive shall be furnished in the following conditions, as specified

Condition A - Primary adhesive with room temperature, 24° to 32°C (75° - 90°F), setting secondary adhesive

Condition B - Primary adhesive with intermediate temperature, 32° - 82°C (90° - 180°F), setting secondary adhesive

2 APPLICABLE DOCUMENTS

2 1 Government publications The issues of the following documents, in effect on date of invitation for bids or solicitations for offers, form a part of this specification to the extent specified herein

AMSC N/A FSC 8040

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SPECIFICATIONS

Federal Specifications

QQ-A-250/5	Aluminum Alloy Alclad 2024, Plate and Steel
TT-S-735	Standard Test Fluids, Hydrocarbor
PPP-B-585	Box, Wood, Wirebound
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-C-843	Cushioning Material, Cellulosic
PPP-D-729	Drums, Shipping and Storage, Steel, 55-Gallon (208 liter)
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock, (Container Grade), and Cut Shapes
PPP-P-291	Paperboard, Wrapping and Cushioning
PPP-P-704	Pails, Metal, Shipping Steel, (1 through 12 Gallon)

STANDARDS

Federal Standards

FED-STD-102	Preservation, Packaging and Packing Levels
FED-STD-123	Marking for Shipment (Civilian Agencies)
FED-STD-175	Adhesives, Methods of Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, Handbooks and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U S. Government Printing Office, Washington, D C 20402).

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(Single copies of this specification and other Federal Specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Business Service Centers in Boston, MA, New York, NY, Philadelphia, PA, Washington, DC, Atlanta, GA, Chicago, IL, Kansas City, MO, Fort Worth, TX, Houston, TX, Denver, CO, San Francisco, CA, Los Angeles, CA and Seattle, WA)

(Federal Government activities may obtain copies of Federal Standardization documents and the index of Federal Specifications, Standards and Commercial Item Description from established distribution points in their agencies)

Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Other publications The following documents from a part of this specification to the extent specified herein Unless a specific issue is identified, the issue in effect on date of invitation for bid or request for proposal shall apply

American Society for Testing and Materials (ASTM) Standards

D2016	Test Methods for Moisture Content of Wood

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

2 3 Order of precedence In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence

3 REQUIREMENTS

3 1 First article Unless otherwise specified in the contract or order (See 4.3.1), the adhesive furnished to this specification shall have been inspected and passed the first article inspection specified herein (See 4.3).

3.2 Materials The adhesive shall consist of a resin or resins furnished as a liquid with or without hardener, or in solid form as stick, powder or film.

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3 2 1 Liquid form Liquid adhesives shall mix readily to a smooth solution or suspension of a consistency suitable for application and shall be free of lumps The components shall not settle out during a normal 8-hour working day

3.2 1 1 Working life The liquid adhesive shall have a minimum working life of 2-1/2 hours at 23° ± 1°C (73 5° ± 2°F) when determined as specified in 4 5 6

3 2 2 Film form Adhesives in the film form shall consist either entirely of adhesive or a carrier impregnated with adhesive The material and treatment of the carrier shall be designated If necessary, an activating solvent shall be furnished and used with adhesive in film form

3 2 3 Accelerator, hardener or modifier When required, an accelerator, hardener, or modifier shall be supplied in powder or liquid form for mixing with the adhesive

3.2 4 Solubility The adhesives shall be capable of being thinned with water or with commercially available solvents specified by the manufacturer Adhesives, before curing, shall be easily dissolved and removed from spreading equipment

3 3 Application

3.3 1 Type I Adhesives shall be capable of being readily applied to aluminum alloy and to wood surfaces in accordance with the manufacturer's instructions at temperatures between 16° and 38°C (60° and 100°F), at relative humidities of 30 to 75 percent

3 3 2 Type II The primary adhesives shall adhere to aluminum alloys The secondary adhesives shall adhere the wood member to the primary coated metal, resulting in a metal to wood joint The adhesives shall be capable of being readily applied to the surfaces in accordance with the manufacturer's instructions at temperatures between 16° and 38°C (60° and 100°F) at relative humidities of 30 to 75 percent

3 4 Curing The time, temperature and pressure shall be as specified herein, unless otherwise approved by the procuring agency

3 4 1 Type I

3 4 1 1 Curing temperature The adhesives shall require not longer than 20 minutes at a temperature not exceeding 168°C (335°F) at the bonding line If a lower temperature is specified for curing, the curing period may be proportionately longer

3.4.1.2 Curing pressure The pressure required for curing the bonded assembly described in 4 4 1 2 shall not exceed 50 pounds per square inch.

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3 4 2 Type II

3 4 2 1 Curing temperature

3 4 2 1 1 Primary Primary adhesives shall require not longer than 20 minutes at a temperature not exceeding 168°C (335°F) at the adhesive line If a lower temperature is specified for curing, the curing period may be proportionately longer

3 4 2 1 2 Secondary

Condition A - Condition A secondary adhesives shall require not longer than 8 hours to cure when exposed to a temperature of 24° to 32°C (75° to 90°F) measured in the adhesive line

Condition B - Condition B secondary adhesives shall require not longer than 5 hours to cure when exposed to a temperature of 32° to 82°C (90° to 180°F) measured in the adhesive line

3 4 2 2 Curing pressure The pressure required for curing the bonded assembly described in 4 4 1 2 shall not exceed 50 pounds per square inch

3 5 Shear strength Shear strength values of specimens tested in 4.4 1.2 shall be as specified in Table I

3 6 Storage life

3 6 1 Type I The adhesive as furnished by the manufacturer shall be capable of meeting Type I requirements for Initial Shear Strength, Shear Strength after Immersion in Salt Water, and Working Life after being stored for 3 months from date of manufacturer at a temperature of 25° ± 1°C (77° ± 2F), or 6 months from date of manufacture at a temperature between 5° and 10°C (41° and 50°F) in airtight containers.

3 6 2 Type II The primary adhesive as furnished by the manufacturer shall be capable of meeting the Type I requirements for Initial Shear Strength, Shear Strength after Immersion in Salt Water, and Working Life after being stored for 3 months from date of manufacture at a temperature of 25 ± 1°C (77° ± 2°F), or 6 months from date of manufacture at a temperature between 5° and 10°C (41° and 50°F) in airtight containers The condition A secondary adhesive as furnished by the manufacturer shall be capable of meeting the Type II requirements for Initial Shear Strength, Shear Strength after Immersion in Salt Water, and Working Life after being stored for 1 year from date of manufacture at a temperature of 25° ± 1°C (77° ± 2°F) in airtight containers The Condition B adhesives as furnished by the manufacturer shall be capable of meeting the Type II requirements for Initial Shear Strength, Shear Strength after Immersion in Salt Water, and Working Life after being stored for 6 months from date of manufacture at a temperature of 25° ± 1°C (77° ± 2°F) in airtight containers.

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TABLE I Shear Strength Values

CONDITION	TYPE I	TYPE II
Initial Shear Strength at 25°± 1°C (77°± 2°F)	900	900
Shear Strength at 82°± 1°C (180°± 2°F)	700	700
Shear Strength after immersion in fresh water for 7 days at 25°± 1°C (77°± 2°F)	700	700
Shear Strength after immersion in salt water for 7 days at 25°± 1°C (77°± 2°F)	700	700
Shear Strength after immersion in hydrocarbon test fluid for 7 days at 25°± 1°C (77°± 2°F)	700	700

1/ minimum values

3 7 Instruction sheet The manufacturer shall provide an instruction sheet with each shipment of the adhesive outlining instructions for its use The instruction sheet shall include the following

(a) The general chemical type of the base resin used in the adhesive

(b) Maximum usable storage life of the adhesive, with and without the addition of accelerators, at various temperatures between 41°F and 100°F

(c) Mixing instructions including recommended type and percentage of accelerator, maximum percent of accelerator allowed, and temperature control necessary during mixing.

(d) Complete recommended metal cleaning processes and treatments including proportions of materials necessary to prepare cleaning media

(e) Application instructions including spread method, number of coats, spread rate and film thickness range

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(f) Drying time between coats and after last coat If a force dry is required, the time and temperature shall be stated

(g) Maximum allowable life of adhesive coated metal prior to assembly and cure

(h) Curing cycle including recommended time, temperature and pressure for each assembly

(i) Any other pertinent information relative to the use and storage of the adhesive

3 8 pH Value - secondary adhesives (Type II) The set secondary adhesive film shall develop a pH value of not less than 3 5 nor more than 11 0 when determined as specified in 4 5 5

3 9 Workmanship Workmanship shall be in accordance with high-grade practice covering this class of material

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.2 Classification of inspections The inspection requirements specified herein are classified as follows

(a) First Article Inspection (See 4 3)
(b) Quality Conformance Inspection (See 4 4)

4 2.1 Inspection Conditions Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the test method or the applicable paragraph of this specification.

4.3 First Article Inspection First article inspection shall be performed by an activity designated by the contracting officer. Inspection shall consist of all the tests and inspections of this specification

4.3.1 Prior approval When a contractor has delivered an acceptable product to this specification, the contracting officer may at his discretion waive the first article requirements for a period of up to three years The contractor must certify that his current product is identical to the previously acceptable product.

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4 3 2 First article sample The first article sample shall consist of 1 quart of adhesive of each type along with necessary hardness or accelerators 1 quart of Secondary adhesive shall accompany each quart of Type II adhesive

4 3 2 1 Manufacturer's certification The contractor shall furnish, with the first article sample, certification that the adhesive meets the storage requirements specified herein (See 3 6)

4 3 2 2 Manufacturer's instructions Two copies of the mixing, application and curing instructions for the adhesive shall also be furnished for approval with the first article test samples (See 3 7)

4 4 Quality conformance inspection

4 4 1 Lot formation Unless otherwise specified, a lot shall consist of all the adhesive of one type and secondary adhesive prepared from one production run and offered for inspection at the same time

4 4 1 1 Source inspection Unless otherwise specified, all materials furnished to this specification shall be source inspected before shipping to the Government

4 4 2 Sampling and inspection

4 4 2 1 Physical property examination Sufficient adhesive shall be randomly selected from each lot to prepare specimens for initial shear strength and shear after immersion in fresh water Failure of any specimen to conform to the requirements of this specification shall be cause to reject the lot representative by the sample

4 4 2 2 Packaging inspection

4 4 2 2 1 Examination for packaging and marking An examination shall be made to determine that packaging and marking comply with the requirements of Section 5 herein Defects shall be scored in accordance with Table II The sample unit for this examination shall be one shipping container fully prepared for delivery except that it shall not be palletized and need not be sealed Shipping containers fully prepared for delivery shall be examined for closure defects The lot size shall be the number of shipping containers in the lot The samples for this examination shall be selected in accordance with MIL-STD-105, Inspection Level S-2 and an AQL of 4.0 defects per hundred units

4 4 2.2 2 Examination for palletization An examination shall be made to determine that the palletization complies with the requirements of Section 5 of this specification Defects shall be scored in accordance with Table III The sample unit shall be one palletized unit load fully prepared for delivery The lot size shall be the number of palletized units in the inspection lot The samples for this examination shall be selected at random in accordance with MIL-STD-105, Inspection Level S-1 and an AQL of 6.5 defects per hundred units

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TABLE II Packaging inspection

EXAMINE	DEFECT
Packaging	Container not as specified, closures not accomplished by specified or required methods or materials Puncture, break, leakage or seepage of contents Non conforming component, component missing, damaged or otherwise defective Bulged or distorted container
Markings	Date omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements

TABLE III Palletization inspection

EXAMINE	DEFECT
Finished dimension	Length, width, or height exceeds specified maximum requirements
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not banded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect, illegible, of improper size, location sequence or method of application

4.5 Specimen preparation

4.5 1 Mixing and thinning of adhesive. The adhesive shall be mixed and thinned, if necessary, in accordance with the manufacturer's instructions

4 5.2 Shear specimens The test specimens shall conform to the form and dimensions shown in Method 1032 of Fed Test Method Std. No. 175 The test specimens shall have a one-sixteenth of an inch thick yellow birch veneer core and 0.032 inch thick faces of aluminum clad aluminum alloy conforming to the requirements of QQ-A-250/5 The birch veneer shall have a moisture content of 6 to 12 percent determined in accordance with ASTM D2016 The shear specimens shall be prepared as specified in Table IV.

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TABLE IV Shear specimen preparation

	Type I adhesives	Type II adhesives
1	Twenty pieces of aluminum alloy shall be cut to the dimensions 4 by 12 inches	
2	Ten pieces of yellow birch veneer shall be cut to the same dimensions, with grain running parallel to the short dimensions	
3	The aluminum-alloy pieces shall be cleaned in accordance with the manufacturer's instructions	
4	The adhesive shall be applied to both surfaces of the birch core and to the one contacting surface of each metal face or as specified in the approved manufacturer's instructions The adhesives shall then be air-dried and precured in accordance with the manufacturer's instructions	4 The primary adhesive shall be applied to one side of the aluminum-alloy sheets, air-dried, precured, and cured in accordance with the approved instructions of the manufacturer
5	Not applicable to Type I See step 6	5 The secondary adhesive shall be applied to both contacting surfaces of the veneer and to the cured primary adhesive -coated aluminum-alloy sheet (The amount of secondary adhesive applied to each surface is usually between 6 and 7 5 grams This is based on 40-50 lbs /1,000 sq ft of single adhesive line)
6	The panels shall be assembled by placing one yellow birch veneer core between two aluminum-alloy sheets so that the adhesive-coated surfaces of both the birch core and the aluminum alloy-sheets are in contact Type II, condition A, adhesive test specimens shall be thus assembled within the maximum open assembly time specified by the manufacturer	
7	The assembly shall be placed in a heated press and cured at the temperature and for the time interval specified by the manufacturer. The pressure applied shall not exceed 50 pounds per square inch.	7. A pressure not exceeding 50 pounds per square inch shall be allowed to remain under pressure for 8 to 16 hours at 25° ± 1°C (77°± 2°F) For condition B, the assembly shall be cured under pressure at the temperature and for the interval specified by the manufacturer in his written instructions The assembly shall be allowed to cool to room temperature under

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pressure. Conditions A and B assemblies shall be allowed to condition for 6 to 7 days at 25°± 1C (77°± 2°F) before cutting into specimens

8 After the specified curing cycle, the bonded panels shall be removed and allowed to cool to room temperature

9 The 10 panels shall be (a) cut and, (b) numbered 1 through 100 according to Method 1032 of Fed Test Method Std No 175

4 6 Test methods. Test methods shall be identified for both types of adhesive, except where otherwise indicated

4 6 1 Examination of product. The adhesive shall be examined to determine conformance with the requirements of this specification with respect to materials and workmanship

4 6 2 Shear strength of bonded joints. Tests shall be conducted in accordance with the methods prescribed in Fed Test Method Std No 175 If the wood utilized for the test is not capable of yielding the minimum strengths specified, a new lot of wood shall be selected for tests The specimens shall be tested as follows

4 6 2 1 Initial shear strength. Specimens numbered 1, 6, 11, 16, etc , shall be placed in distilled water at 25°± 1°C (77°± 2°F) and allowed to remain therein for 1 hour They shall then be allowed to dry overnight at 24° to 27°C (75° to 80°F) and a relative humidity of 50 to 70 percent and tested in the dry state

4 6 2 2 Shear strength at elevated temperature. Specimens numbered 2, 7, 12, 17, etc , shall be conditioned at 82°± 1°C (180°± 2°F) for 1 hour and shall then be tested in a heat chamber at that temperature

4 6 2 3 Shear strength after immersion in fresh water. Specimens numbered 3, 8, 13, 18, etc , shall be placed in distilled water for 7 days at 25°± 1°C (77°± 2°F) then be tested in the wet state

4 6.2 4 Shear strength after immersion in salt water. Specimens numbered 4, 9, 14, 19, etc , shall be immersed in a 3 percent salt water solution for 7 days at 25°± 1°C (77°± 2°F) then be tested in the wet state.

4.6 2 5 Shear strength after immersion in hydrocarbon test fluid. Specimens numbered 5, 10, 15, 20, etc , shall be immersed in hydrocarbon test fluid conforming to TT-S-735, Type II, for 7 days at 25°± 1°C (77°± 2°F) then be tested immediately after being removed from the test fluid

4.6 3 Storage life.

4.6 3 1 Type I adhesive. A suitable amount of the Type I adhesive being tested shall be stored under the conditions specified in 3 6 1. At the end of the specified storage period, the adhesive shall be capable of meeting the

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Type I requirements of this specification for Working Life, Initial Shear and Shear Strength after Immersion in Salt Water (10 specimens each)

4 6 3 2 Type II adhesives A suitable amount of the Type II primary adhesive and the designated secondary adhesive shall be stored under the conditions specified in 3 6 2 After the specified time interval has passed, the adhesive combination shall meet the Type II requirements of this specification for Working Life, Initial Shear and Shear Strength after Immersion in Salt Water (10 specimens each)

4 6 4 Solubility The spreading equipment (brushes, etc) shall be washed in the specified solvent and examination made for evidence of nonsolubility

4 6 5 pH value The adhesive shall be prepared and tested in accordance with Method 4011 of Fed Test Method Std No 175.

4 6 6 Working life Pour approximately 200 grams of freshly mixed adhesive into a large glass test tube or beaker approximately 35 mm (1 inch) in diameter, provided with a glass stirring rod, and immerse the tube with contents in a water bath held at 23 1°± 1°C (73 5°± 2°F) Stir the adhesive frequently with the rod, and note the time at which the adhesive is considered to be definitely no longer spreadable on a wood surface with stiff bristle brush The time to reach this consistency after mixing is the liquid working life of the adhesive at 23 1°± 1°C (73 5°± 2°F)

5 PACKAGING

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with FED-STD-102

5 1 Preservation Preservation shall be Level A, B, or Commercial as specified (see 6 5)

5 1 1 Level A

5 1 1 1 Unit packaging Unless otherwise specified, all adhesives and accelerators, shall be furnished in cans, pails, or drums If packaged in cans and an accelerator or a two-part adhesive is required, the metal containers for both the adhesive or adhesives, and the equivalent proportions of accelerator required shall be included in a unit fiberboard box, the quantities of the adhesive or adhesives and accelerator so packaged, when mixed according to the manufacturer's instructions, shall produce the specified quantity of adhesive. The fiberboard boxes shall conform to PPP-B-636, Class Weather-Resistant If adhesive is packaged in pails or drums, it will be necessary that the pails or drums, with their proportionate additional containers of adhesive or accelerator, if applicable, be packaged together. This may be accomplished through the utilization of multiple compartment metal containers or suitable modified single compartment containers. When shipment is to be made in a modified single compartment container, it will be necessary to submit the proposed container or a drawing of the same to the procuring agency, prior to shipment The capacity of all adhesive containers shall be as specified by the procuring agency The metal containers shall be in accordance with the following requirements, as applicable.

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5.1.1.2 Liquid or powdered materials Liquid, or powdered materials shall be packaged in Type V, Class 2, cans conforming to PPP-C-96, in Type II 5-gallon pails with lug closures conforming to PPP-P-704, or in Type IV 55-gallon open top style metal drums with bolted ring seal closures conforming to PPP-D-729

5 1.1 3 Cushioning Unless otherwise specified by the procuring agency, when unit fiberboard boxes are required, the contents shall be secured snugly by means of vertical separators and cushioning Vertical separators shall be made of fiberboard, Type CF, Class Domestic, Variety SW or DW of PPP-F-320 Cushioning shall be flexible, corrugated paper, Type I, Style 2 of PPP-P-291, or cellulosic cushioning Type II, Class A, of PPP-C-843

5 1 2 Level B The adhesive shall be packaged as specified in 5 1 1, except the PPP-B-636 box shall be Class Domestic

5.1 3 Level C The adhesive and accelerators shall be packaged in accordance with the supplier's commercial practice

5 2 Packing Packing shall be Level A, B or Commercial as specified (see 6.5)

5.2 1 Level A Type I or II adhesive or accelerators shall be packed in close-fitting boxes conforming to PPP-B-585, Class 3, PPP-B-591, Class II, PPP-B-601, Type III, PPP-B-621, Class 2, PPP-B-636, V3c or PPP-B-640, Class 2, Grade A The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed 200 pounds The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification No packing is required for the pails and drums

5.2 2 Level B Type I or II adhesive or accelerators shall be packed in close-fitting boxes conforming to PPP-B-585, Class 1, PPP-B-591, Class 1, PPP-B-601, Type IV, PPP-B-636, Class Domestic or PPP-B-640, Class 1, Grade A The boxes shall be closed in accordance with the applicable specification or appendix thereto The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed approximately 150 pounds The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification No packing is required for the pails and drums

5 2 3 Level C Type I or II adhesive or accelerators shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation

5.3 Unitization. When specified in the contract or order the shipping container shall be palletized in accordance with MIL-STD-147.

5.4 Marking

5.4.1 Civil agencies In addition to markings required by the contract or order, the cans, pails and drums shall be marked in accordance with FED-STD-123.

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5 4 2 Military activities In addition to markings required by the contract or order, the cans, pails and drums shall be marked in accordance with MIL-STD-129

6 NOTES

6.1 Intended use

6 1 1 Type I adhesives The Type I adhesive covered by this specification is intended primarily for use in the structural bonding of aluminum alloy to wood It shall not be used in aircraft for the fabrication of primary structure or for adhering wood to any metal other than aluminum alloy without the specific approval of the procuring agency

6 1 2 Type II adhesives Type II adhesives covered by this specification are intended primarily for use in fabricating aluminum-wood-aluminum structural sandwich panel constructions, particularly where low density wood cores are used, as well as for the structural bonding of aluminum to wood attachments not of sandwich type construction It shall not be used in aircraft for the fabrication of primary structure or for adhering wood to any metal other than aluminum alloy without specific approval of the procuring agency. Type II, condition A adhesive is intended for applications where small areas are involved or where heat cannot be used in the final assembly operation Type II, condition B adhesive is interded or use in cases where the fabrication of large items necessitate a final assembly time exceeding that permissible with the condition A adhesive

6 2 Ordering data

6 2 1 Acquisition requirements. Acquisition requirements should specify the following

(a) Title, number and date of this specification.

(b) Type and condition of adhesive (See 1 2), adhesive code number (see 3 2 4 1)

(c) Capacity and kind of container and quantity desired

(d) Level of packaging and packing required (see 5 1 and 5 2)

(e) Palletization, when required (see 5.3)

(f) Whether first article inspection is required (see 4 3 1)

6.3 First article When a first article inspection is required, the item should be a standard production item from the contractor's current inventory. The Contracting Officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first articles test results and disposition of first articles (see 4.3) Invitation for bids should provide that the Government reserves the right to waive the requirements for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government (see

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4.3 1), and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract

6 4 General In the design of parts and assemblies requiring the use of adhesives covered by this specification, consideration should be given to the heat and pressure required during the bonding operation. The curing of the bonded joint in the case of Type I bonding, or the prime coat in the case of Type II bonding, may be expected to require a curing temperature as high as 325F The curing pressure of 50 psi specified in Section 3 is a necessary requirement for determining the suitability of an adhesive or adhesive combination for general production use However, shop facilities permitting higher pressures in accordance with the manufacturer's instructions may be used whenever practicable

6 5 Storage Wherever practicable, adhesive conforming to this specification should be stored at a temperature of 5 to 10C (41 to 50F) If transit and normal storage conditions necessitate exposure of the adhesive for prolonged periods to temperatures in excess of 80F, it can be expected, in general, that the storage life of the adhesive will be proportionally reduced

6.6 Subject term (key word) listing

Adhesive
Aluminum to wood
Structural

6 7 Asterisks are not used in this revision due to the extensiveness of the changes

Military interests

Custodians
Army - MR
Navy - AS
Air Force - 99

User activities
Army - CR
Navy - MC

Preparing activity
Navy - AS
(Project No 8040-0467)

Civilian agencies interest.
GSA - FSS

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